YAKEEN

NEET 2.0 2027

Physics | Lecture - 4

Basic Maths and Calculus

By- Manish Raj (MR Sir)



Todays Goal

→ A/P, G/P & straight line graph.

H/W

Phasor diagram

$\theta = \omega t$

(#) $Y_1 = A \sin(\omega t) \Rightarrow \phi_1 = \omega t$

(#) $Y_2 = A \sin(\omega t + 60^\circ) \Rightarrow \phi_2 = \omega t + 60^\circ$

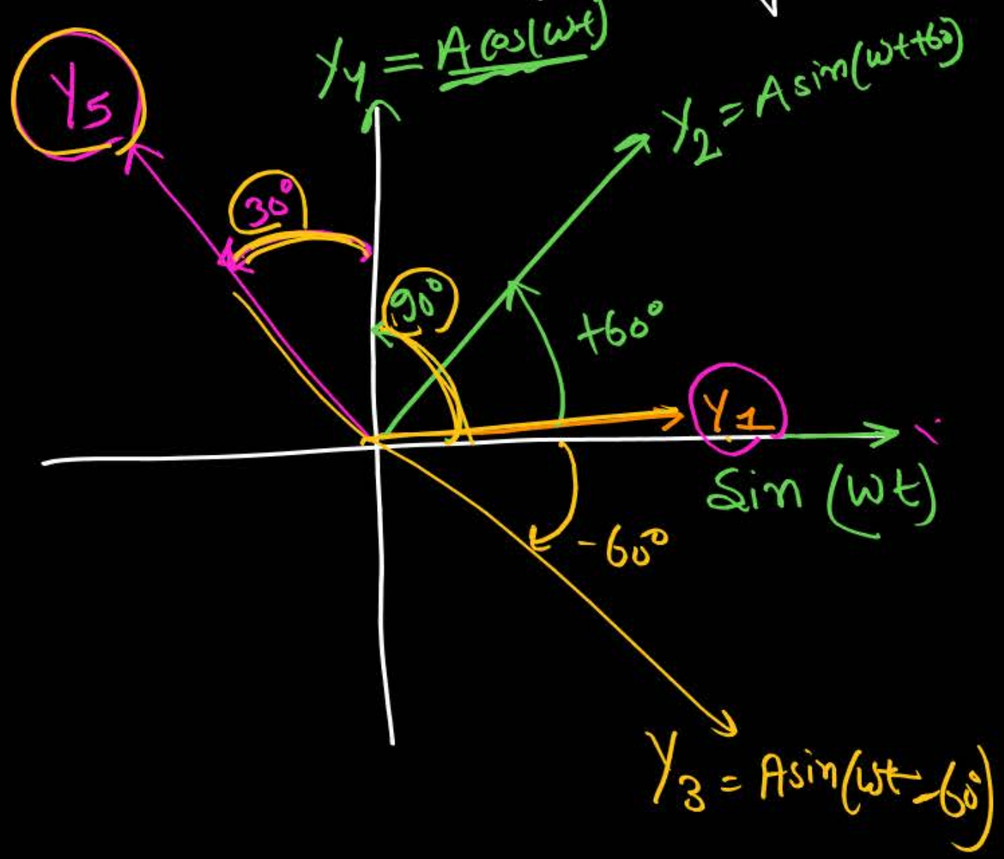
(#) $Y_3 = A \sin(\omega t - 60^\circ) \Rightarrow \phi_3 = \omega t - 60^\circ$

(#) $Y_4 = A \sin(\omega t + 90^\circ) = +A \cos(\omega t)$

(#) $Y_5 = A \cos(\omega t + 30^\circ)$

Phase diff^n B/w Y_5 & Y_1 $\Delta\phi = 120^\circ$ ✓

Phasor diagram (Not graph)



(6) find Phase diff'

ϕ_1

$Y_1 = \sin(\omega t + 30)$ आगे

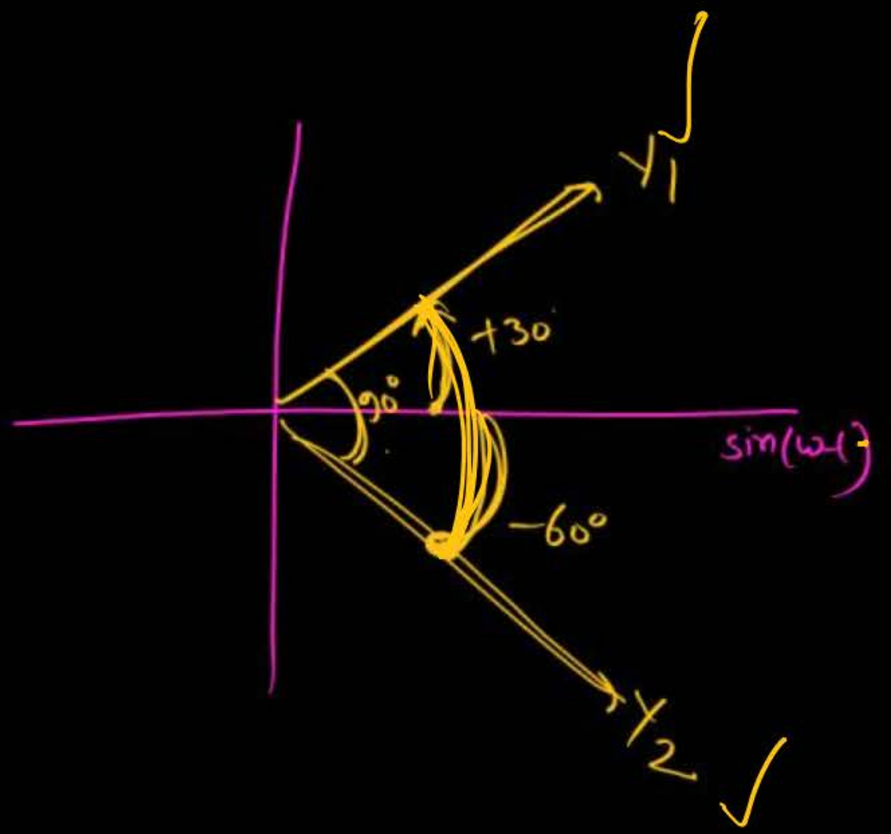
$Y_2 = \sin(\omega t - 60)$ (पीछे)

$\Delta\phi = \phi_1 - \phi_2$

$= (\omega t + 30) - (\omega t - 60)$

$= \omega t + 30 - \omega t + 60$

$= 90°$



(Q) find phase difference b/w Y_2 and X_1

$$X_1 = A\cos(\omega t - \pi/2)$$

$$Y_2 = A\cos(\omega t + \pi/3)$$

$\pi/3 = 60°$

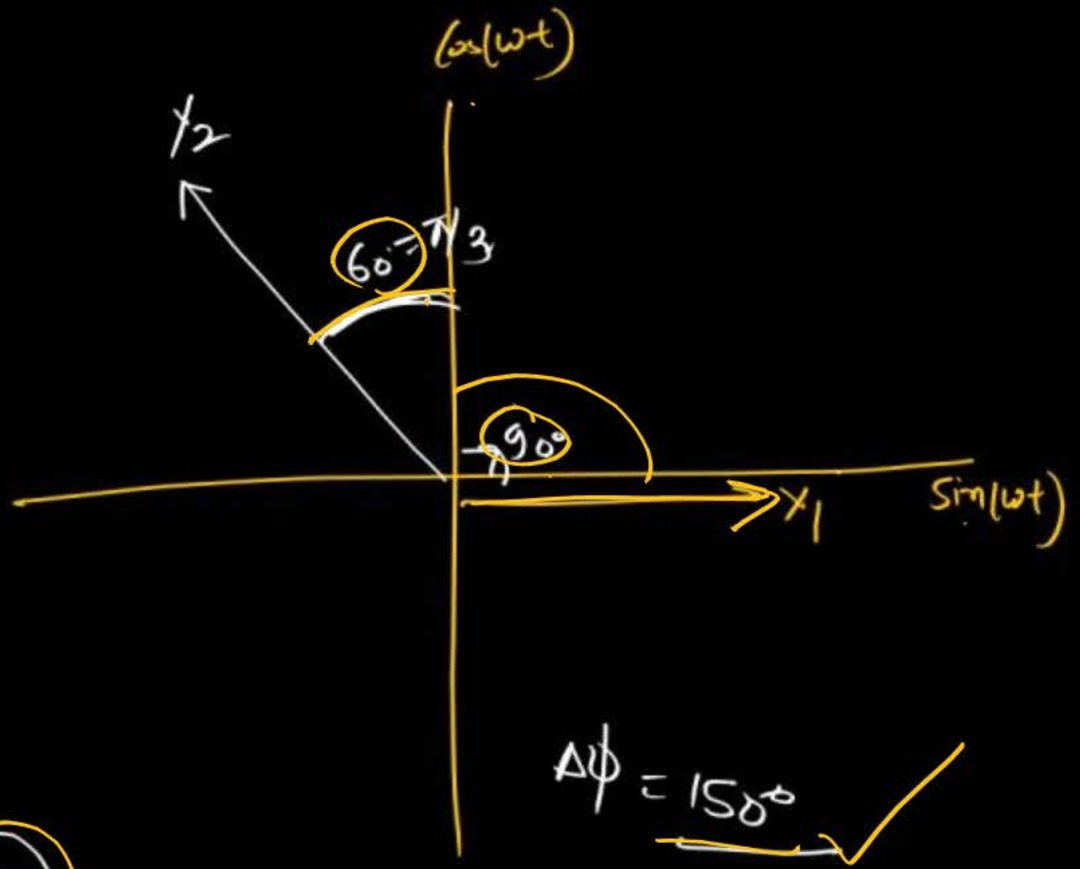
ϕ_2

Soln

$$\Delta\phi = \phi_2 - \phi_1$$

$$= \omega t + \pi/3 - \omega t + \pi/2$$

$$= \frac{\pi}{3} + \frac{\pi}{2} = \frac{5\pi}{6} = \frac{5}{6} \times 180^{30} = 150°$$



$\Delta\phi = 150°$

find phase difference b/w Y_1 & Y_2

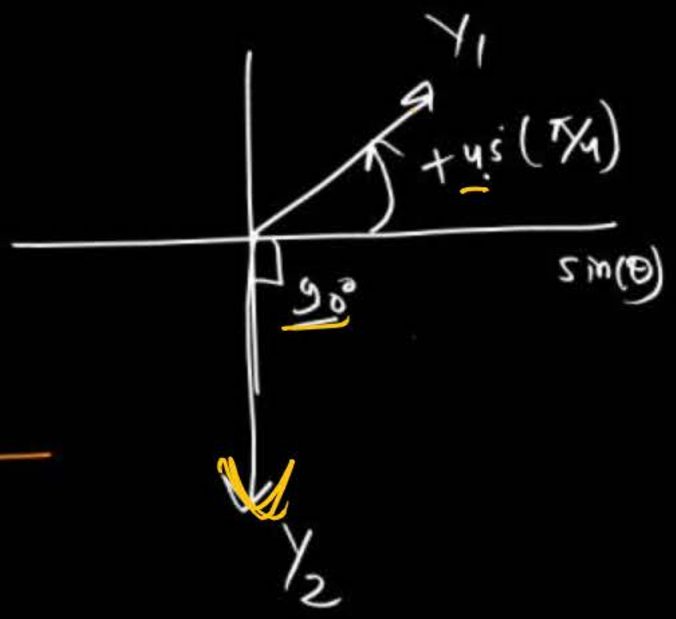
① $Y_1 = 4\sin(\theta + \pi/4)$

$Y_2 = 3\sin(\theta - \pi/2)$

$\Delta\phi = \phi_1 - \phi_2$

$= \theta + \pi/4 - \theta + \pi/2$

$= \frac{\pi}{4} + \frac{\pi}{3} = 135°$



② $Y_1 = 4\sin(\theta + \pi/3)$

$Y_2 = 3\sin(\theta + \pi/6)$

find Phase difference

$Y_1 = 3\cos(\omega t - 90^\circ) = 3[\cos(\omega t)\cdot\cos 90^\circ + \sin 90^\circ \sin(\omega t)]$

$= 3(0 + \sin\omega t) = 3\sin(\omega t)$

$Y_2 = 4\sin(\omega t)$

$\Delta\phi = 0$ ✓

Soln $\Delta\phi = \phi_2 - \phi_1$

$= \omega t - (\omega t - 90)$

$= \omega t - \omega t + 90$

$= 90^\circ$ (wrong) ✗

$Y_1 = 3\sin(\omega t)$

$Y_2 = 4\cos(\omega t + \pi/2)$

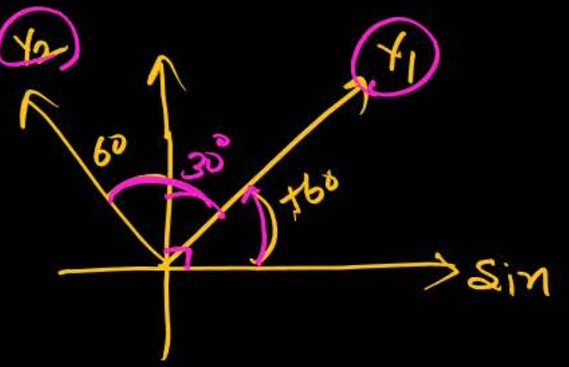
Soln



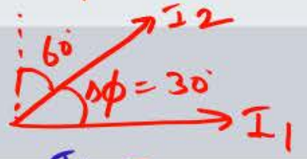
#

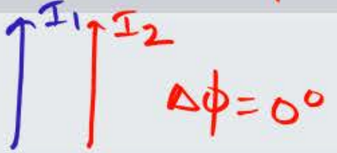
Equation – 1	Equation – 2	Phase difference
$I_1 = I_0 \sin(\theta + \pi/3)$	$I_2 = I_0 \sin(\theta - \pi/6)$	
$I_1 = I_0 \sin(\theta + \pi/3)$	$I_2 = I_0 \cos(\theta + \pi/3)$	
$I_1 = 4 \sin(\omega t)$	$I_2 = I_0\, 4 \cos(\omega t - \pi/3)$	
$I_1 = 3 \sin(\omega t + \pi/2)$	$I_2 = 5 \cos(\omega t)$	
$I_1 = 4 \cos(\omega t - \pi/2)$	$I_2 = 4 \sin(\omega t + \pi/2)$	
$I_1 = 3 \sin(\omega t + \pi/3)$	$I_2 = 3 \cos(\omega t)$	
$I_1 = 4 \sin(\theta - 60°)$	$I_2 = 4 \cos(\theta - 30°)$	



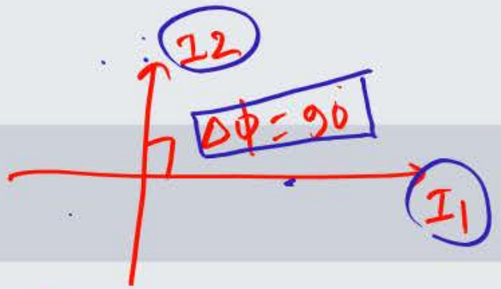


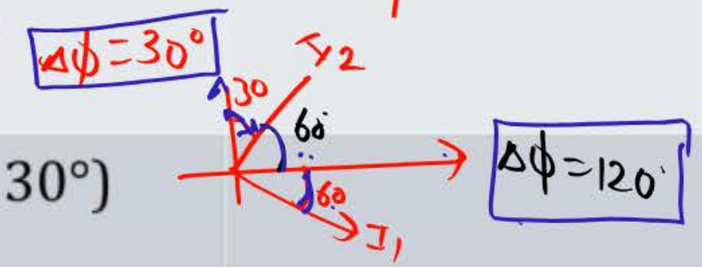
→ $\Delta\phi = 90°$





Equation – 1	Equation – 2	Phase difference
$I_1 = I_0 \sin(\theta + \pi/3)$	$I_2 = I_0 \sin(\theta - \pi/6)$	
$I_1 = I_0 \sin(\theta + \pi/3)$	$I_2 = I_0 \cos(\theta + \pi/3)$	
$I_1 = 4 \sin(\omega t)$	$I_2 = I_0\, 4 \cos(\omega t - \pi/3)$	
$I_1 = 3 \sin(\omega t + \pi/2)$	$I_2 = 5 \cos(\omega t)$	
$I_1 = 4 \cos(\omega t - \pi/2)$	$I_2 = 4 \sin(\omega t + \pi/2)$	
$I_1 = 3 \sin(\omega t + \pi/3)$	$I_2 = 3 \cos(\omega t)$	
$I_1 = 4 \sin(\theta - 60°)$	$I_2 = 4 \cos(\theta - 30°)$	

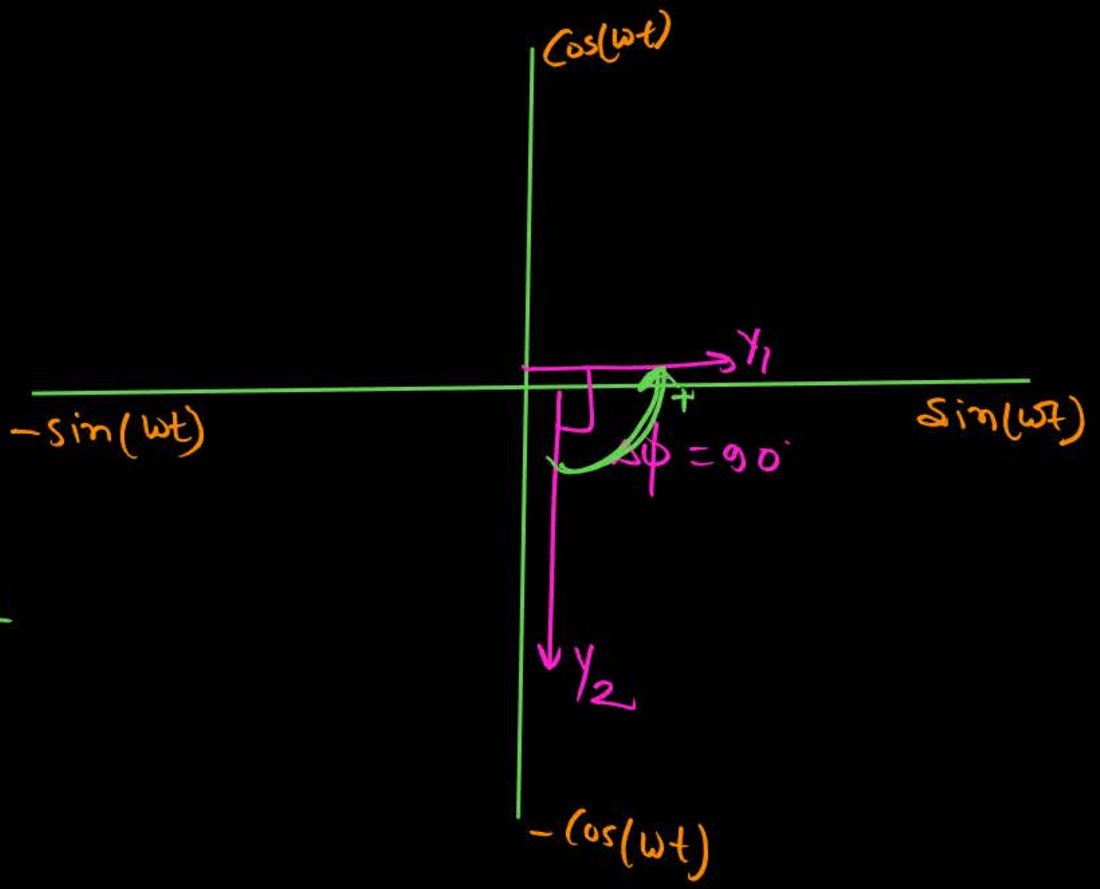




② Two S.H.M Performing Partiue have eqⁿ.

$$Y_1 = 4\sin(\omega t)$$

$$Y_2 = -4\cos(\omega t)$$



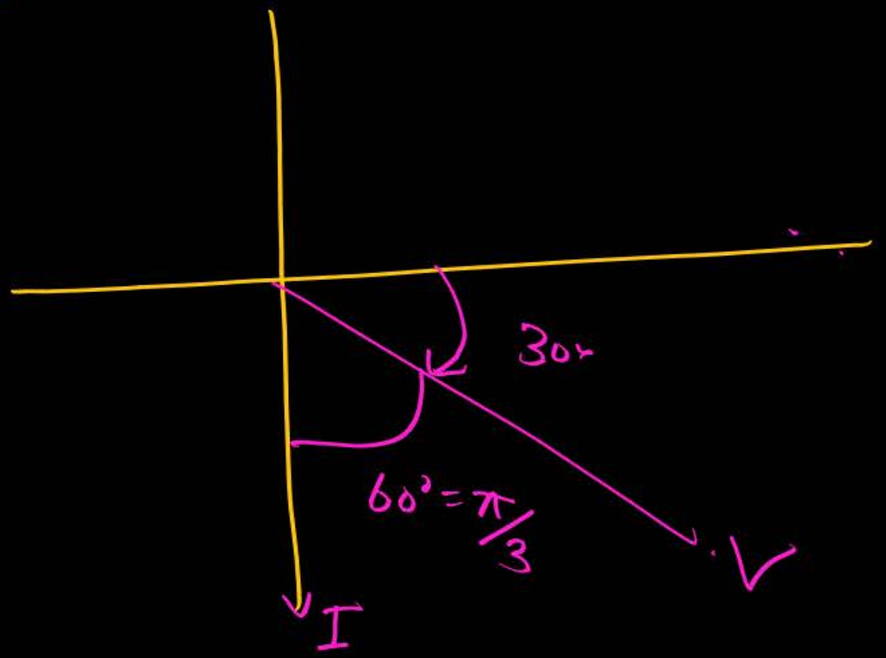
" Y_1 is leading by 90° from Y_2
आगे

Question

Equation of a.c. current $I = -I_0 \cos(\omega t)$ and a/c voltage is $V = V_0 \sin(\omega t - \pi/6)$ then find phase difference between current and voltage:

1. $\frac{2\pi}{3}$
2. $\frac{\pi}{3}$
3. $\frac{\pi}{6}$
4. $\frac{5\pi}{6}$

Soln

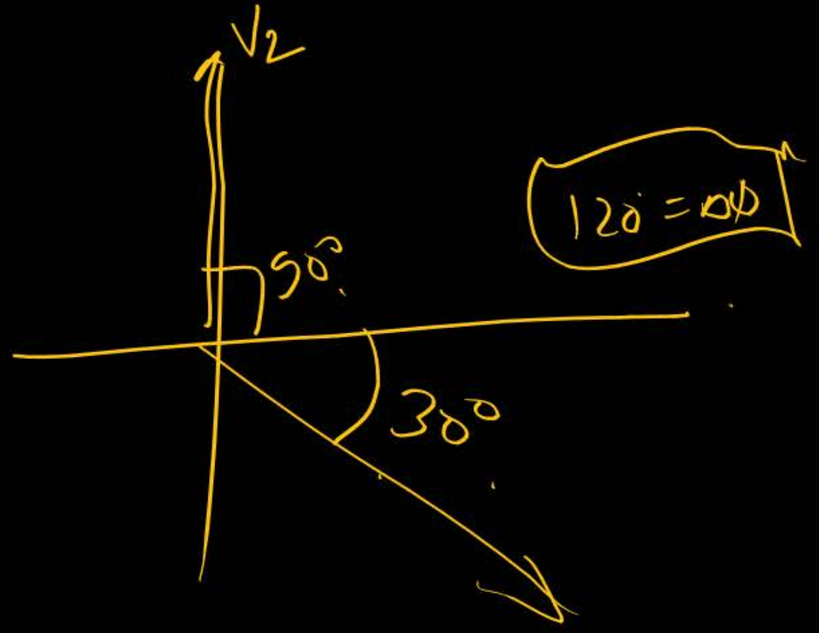


Question No copy

Current in A/C circuit is $I_1 = I_0 \sin(\omega t - 30°)$ and voltage across it $V = V_0 \cos(\omega t)$

Find phase difference?

Soln



Question

Copy

Equation of two oscillating particle

$x_1 = 4\sin(\omega t)$ $x_2 = 3\cos(\omega t)$

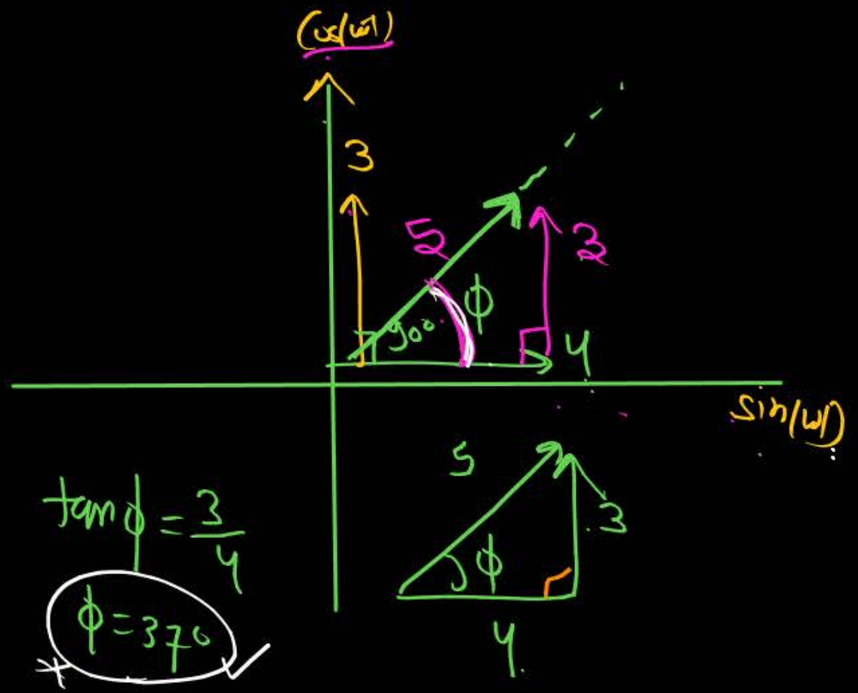
Find equation of ~~super imposed~~ addition SHM

$x = x_1 + x_2$

$x = 4\sin(\omega t) + 3\cos(\omega t)$

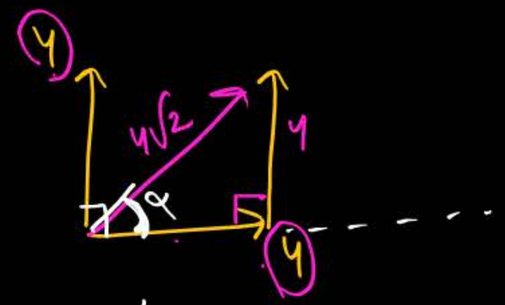
$x = 5\sin(\omega t + \phi)$

$x = 5\sin(\omega t + 37°)$



COPY

(Q) $Y = 4\sin(\omega t) + 4\cos(\omega t) = 4\sqrt{2}\sin(\omega t + 45^\circ)$



$H = \sqrt{4^2 + 4^2}$

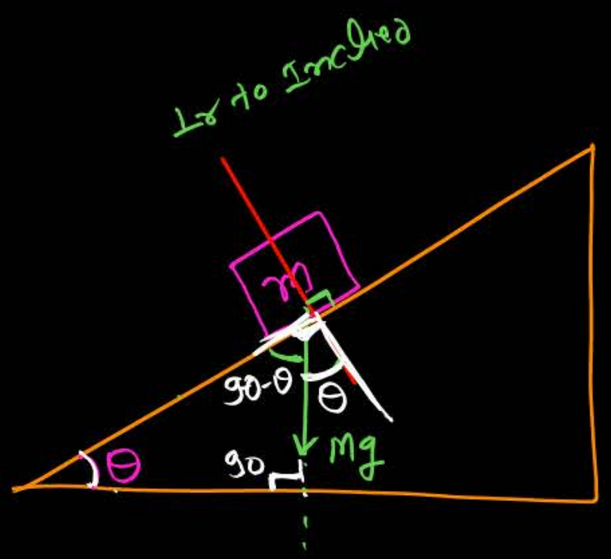
$= \sqrt{16+16}$

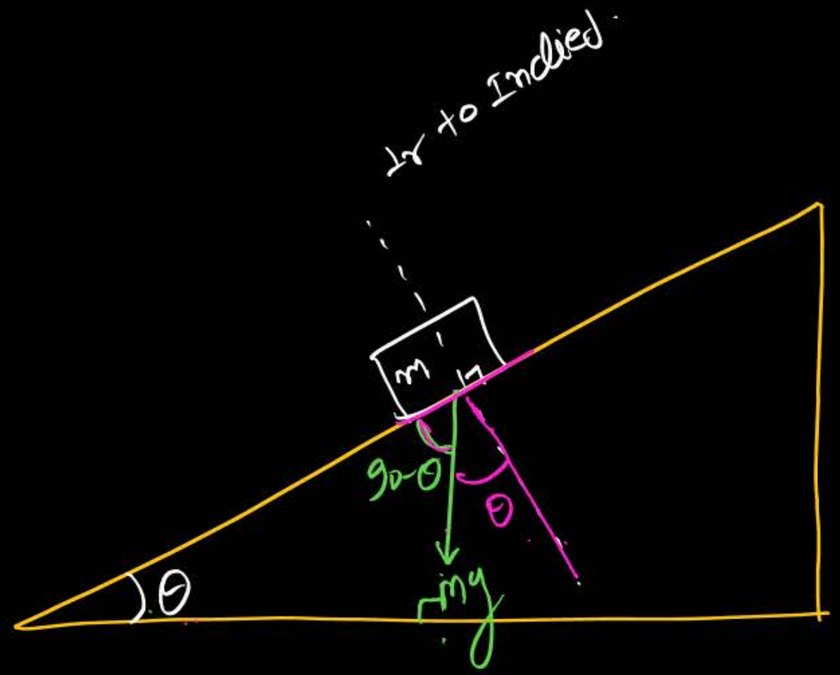
$= \sqrt{32} = \sqrt{16 \times 2}$

$= 4\sqrt{2}$

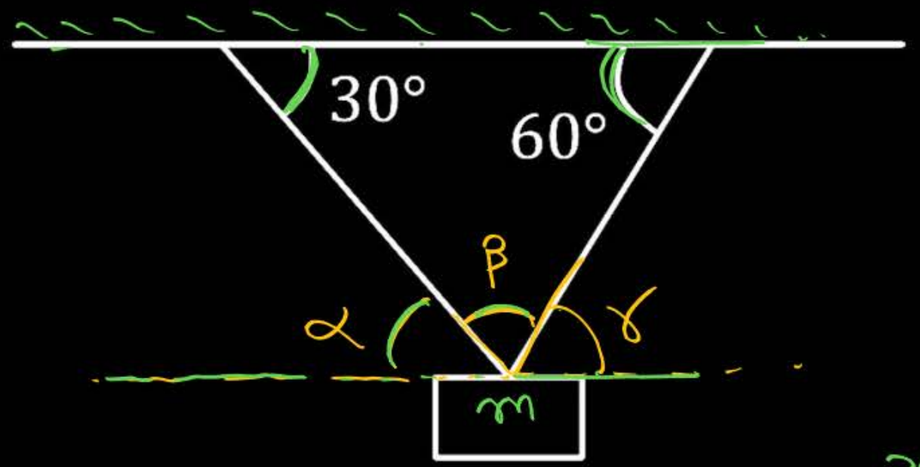
$\tan\theta = \frac{4}{4} = 1$

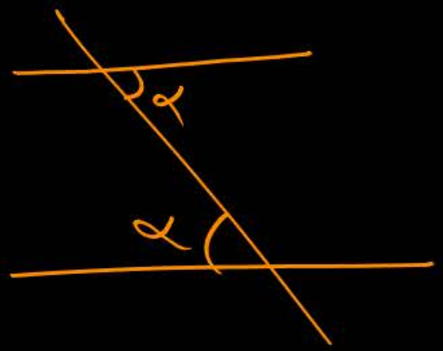
$\theta = 45^\circ$ ✓


⊥r to Inclined
m
90-θ
θ
θ
90
mg


⊥r to Inclied
m
90-θ
θ
θ
mg

find α, β & γ





$30+60+\beta=180$

$\alpha=30°$ $\gamma=60°$

$\beta=180-90=90°$

A/P (Arithmetic Progression) → Sequence of number In which difference b/w any two consecutive terms is always constant.

or

that constant is called Common difference (d)

a , $a+d$, $a+2d$, $a+3d$, $a+4d$, $a+5d$, . $a+(n-1)d$ . . .

a → 1st digit

$a+d$ → 2nd term

$a+2d$ → 3rd term

$a+3d$ → 4th ..

$$d = n^{th}\text{ term} - (n-1)^{th}\text{ term}$$

$$\text{value of } n^{th}\text{ term} = a+(n-1)d$$

Ex $a=7$ $d=3$

7, 10, 13, 16, 19, 22, 25 ...

value of 20th term $= 7+(20-1)3$

$= 7+(19\times3) = 7+57 = 64$

$$\text{Sum of } n\text{-term} = \frac{n}{2}\left[1^{st}\text{ term} + n^{th}\text{ term}\right]$$

$$= \frac{n}{2}\left[a+a+(n-1)d\right]$$

$$= \frac{n}{2}\left[2a+(n-1)d\right]$$

(Q) (GP)

(Q) A/P series 2, 5, 8, 11, 14 ..... find

(i) 1st term $= 2$

(ii) d (common diff^n) $= 11 - 8 = 3$

(iii) value of 10th term $= a + (n-1)d$

$= 2 + (10-1) \times 3 = 2 + 27 = 29$

(iv) sum of 10-term $= \frac{10}{2}[2 + 29] = \left[\frac{1^{st}\text{ term} + n^{th}\text{ term}}{2}\right] \times n$

$= 5[31] = +155$



Question

Find 10th term and sum of 1st 10 term.

7, 11, 15, 19, __, __, __, __, __, __

1st term = 7

$d = 19 - 15 = 4$

10th term $= 7 + (10-1) \times 4$

$= 7 + 9 \times 4$

$= 7 + 36$

$= 43$

$\text{Sum} = \left(\frac{7+43}{2}\right) \times 10 = 250$

Question

Which term of AP 27, 24, 21, ... is zero.

n^{th} term $= 0$

$a = 27$

$d = -3$

n^{th} term $= a + (n-1)d$

$0 = 27 + (n-1) \times (-3)$

$0 = 27 - 3n + 3$

$3n = 30$

$\boxed{n = 10}$

Question No Copy

28, 22, x, y, 4 is an AP then value of x and y is :

$a = 28$

$d = 22 - 28 = -6$

$x = 28 + (3-1)(-6)$

3rd $= 28 - 12$

$= 16$

$Y = 10$

Question

Sum of 1st n-natural number $1 + 2 + 3 + \ldots\ldots\ldots n$.

1. $\frac{n^2}{2}$
2. $\frac{n(n+1)}{2}$
3. $\frac{n(n-1)}{2}$
4. $\frac{n^2(n+1)}{6}$

$1 + 2 + 3 + 4 + 5 + 6 - - - - - n$

$a = 1$
$d = 1$

n^{th} ter $= 1 + (n-1) \times 1$

$= 1 + n - 1$

$= n$

$\text{Sum} = \frac{n}{2}[1 + n]$

$= \frac{n(1+n)}{2}$

If $n = 1$
Sum $= 1$

If $n = 2$
then
Sum $= 3$

Question

copy

25 tuning fork are arranged in increasing order of frequency. Frequency difference is 10 then find frequency of 1st tuning fork if frequency of 25th is 400 Hz.

$d = 10$

$a = ?$

25th term $= 400$ ✓

25th term $= a + (n-1)d$

$400 = a + (25-1) \times 10$

$40 = a + 24 \times 10$

$a = 40 - 240$

$= 160$

2, 6, 18, 54, ...

$a = 2$ $d = 3$

→ Not a.p. ✗

G/P (Geometric Progression) → A sequence in which, Ratio of any two consecutive number is constant, that constant is c/d common ratio(r)

→ Next term found by multiplying Previous number with (constant common ratio.)

a = 1st term r = Common ratio.

↓

$a, ar, ar \times r = ar^2, ar^3, ar^4, ar^5 \ldots$

Value of nth term $= ar^{(n-1)}$

Comm ratio $=$ $\frac{\text{any term}}{\text{Previous term}}$

Sum of n-term (all term) $= \frac{a}{1-(r)}$ if $(r<1)$

(1) 2, 4, 8, 12, 24, 48 ⟶ Not GP

(2) 2, 4, 8, 16, 32, 64 ⟶ GP $a=2$ $r=\frac{32}{16}=2$

No sum

(3) $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}$ ⟶ G/P $a=1$ $r=\frac{1/8}{1/4}=\frac{4}{8_2}=\frac{1}{2}$

$\text{Sum}=\frac{a}{1-r}=\frac{1}{1-\frac{1}{2}}=\frac{1}{1/2}=2$

(4) $1, -\frac{1}{2}, +\frac{1}{4}, -\frac{1}{8}, +\frac{1}{16}, -\frac{1}{32}$ ⟶ G/P $a=1$

$r=\frac{-1/2}{1/4}=-\frac{4}{8}=-\frac{1}{2}$ $\text{Sum}=\frac{1}{1-(-\frac{1}{2})}=\frac{2}{3}$

G/P series

(5) $1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \frac{1}{81}, \ldots$

$a = 1$

$r = \frac{\frac{1}{27}}{\frac{1}{9}} = \frac{9}{27_3} = \frac{1}{3}$

$$\text{Sum} = \frac{1}{1 - \frac{1}{3}} = \frac{1}{\frac{2}{3}} = \frac{3}{2}$$

Question

$V = \frac{kq}{r}$ (hint)

Find potential at origin? (electrostatic repo)

V=??	+q	−q	+q	−q	+q	−q
(0, 0)	(1, 0)	(2, 0)	(4, 0)	(8, 0)	(16, 0)	(32, 0)

Potential at origin

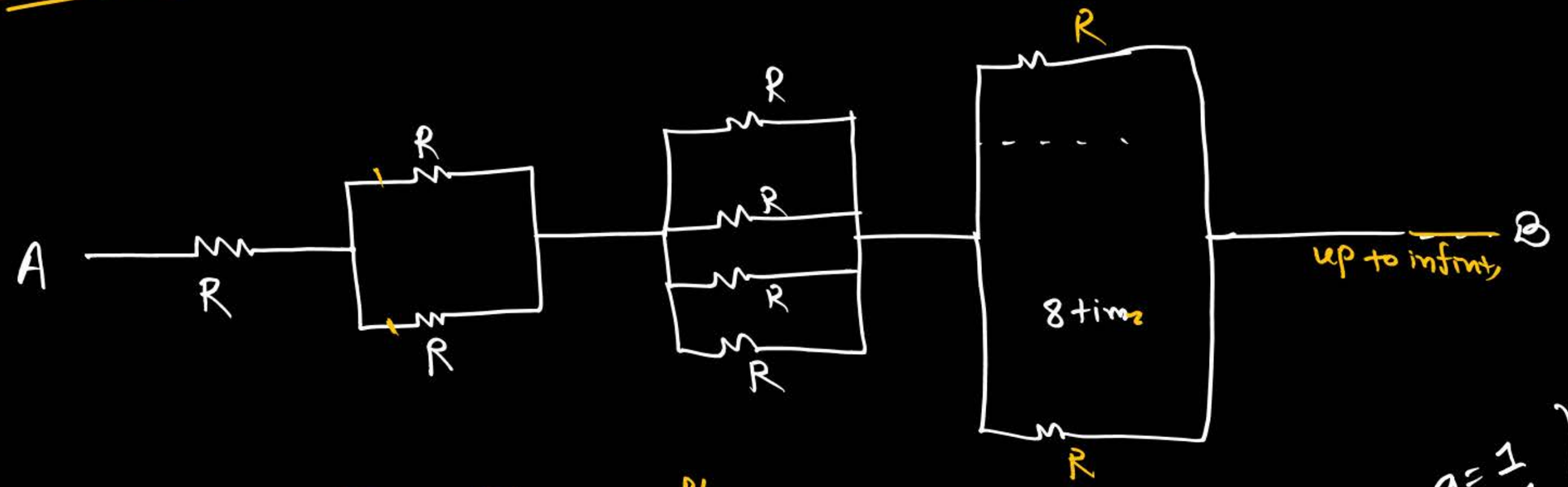
$$V = \frac{kq}{1} - \frac{kq}{2} + \frac{kq}{4} - \frac{kq}{8} + \frac{kq}{16} - \frac{kq}{32} + -$$

$$V = kq\left[1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} + - -\right]$$

G.P

$$= kq\left[\frac{1}{1-\left(-\frac{1}{2}\right)}\right] = kq\left(\frac{1}{1+\frac{1}{2}}\right) = \frac{2kq}{3}$$ Ans

find R_{AB}





GP series $a = 1$, $r = \frac{1}{2}$

$$R_{AB} = R + \frac{R}{2} + \frac{R}{4} + \frac{R}{8} + \cdots = R\left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots\right] = R[2]$$

$$= 2R$$

Question

H/w

Charge q is placed on x-axis of co-ordinate (1, 0), (2, 0), (4, 0), (8, 0) and so on then find force on charge q_0 which is at origin.

Question

HW

5-resistance connected in series, resistance of each resistor is half of previous resistance if 1st resistance of value 10Ω then value of 5th resistance.

Question

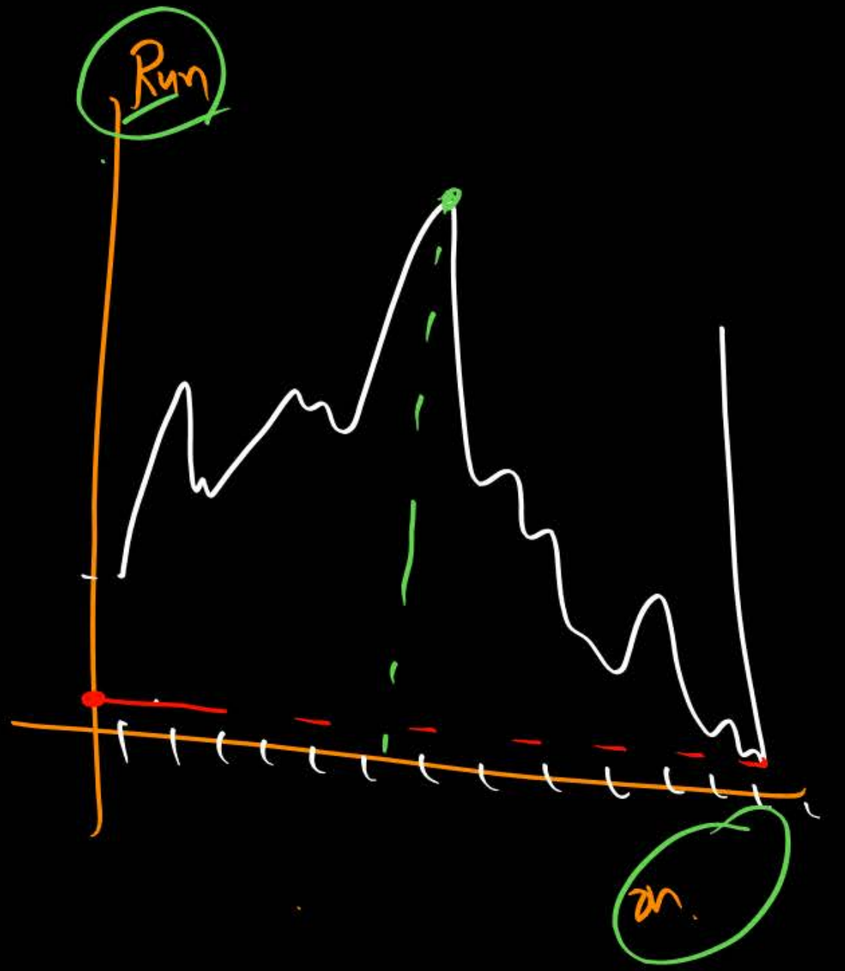
H/W

Find potential at origin?

0	$+q$	$-q$	$+q$	$-q$	$+q$	$-q$
(0, 0)	(1, 0)	(2, 0)	(4, 0)	(8, 0)	(16, 0)	(32, 0)

No copy

1	7	10	8
2	8	11	7
3	6	12	15
4	5	13	14
5	6	14	1
7	3	15	0
8	2	16	9
9	1	17	7
		18	3
		19	6
		20	



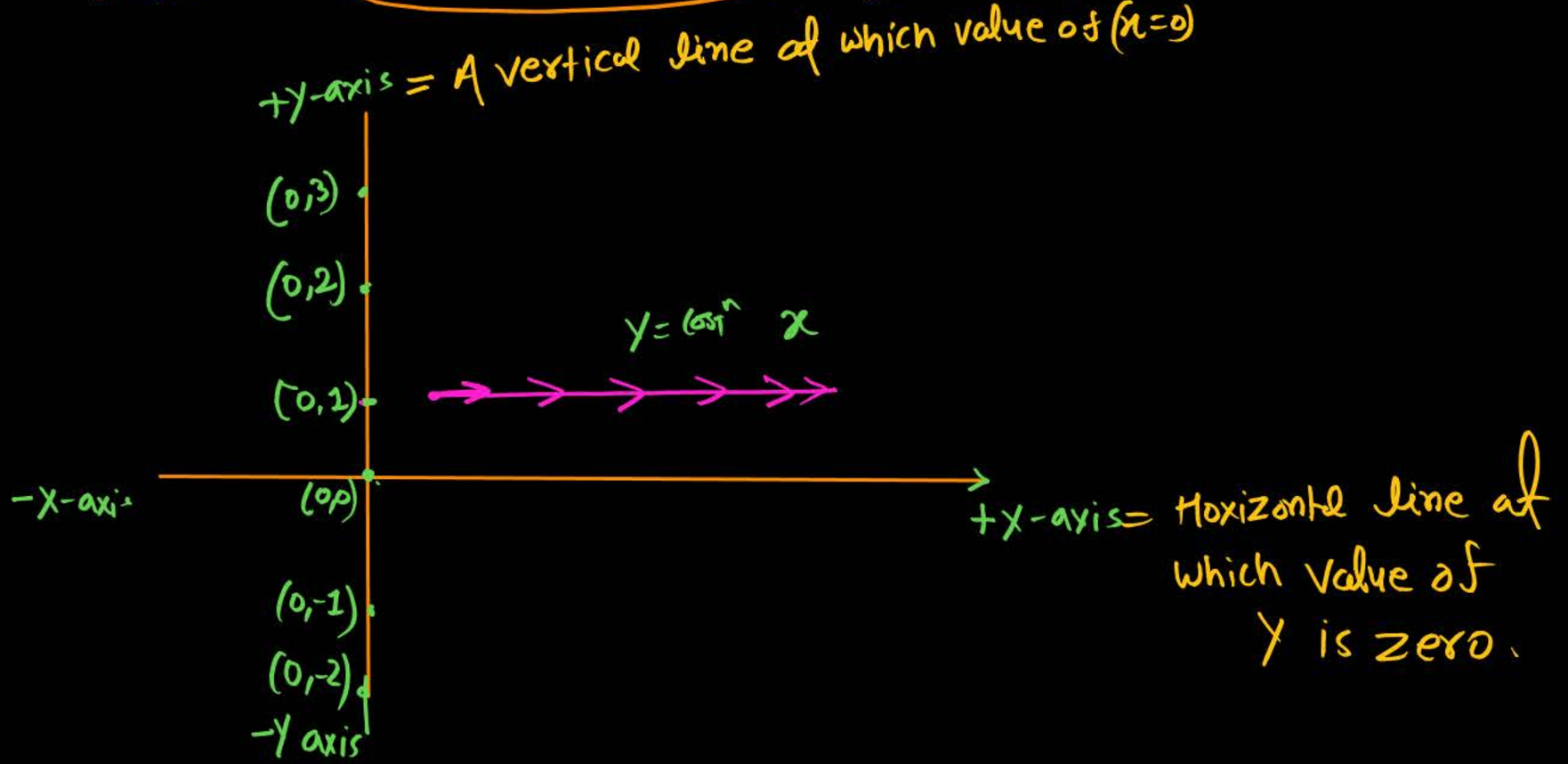
copy

Graph ⟶ A diagram in which a line or curve shows the relation between two physical quantity.

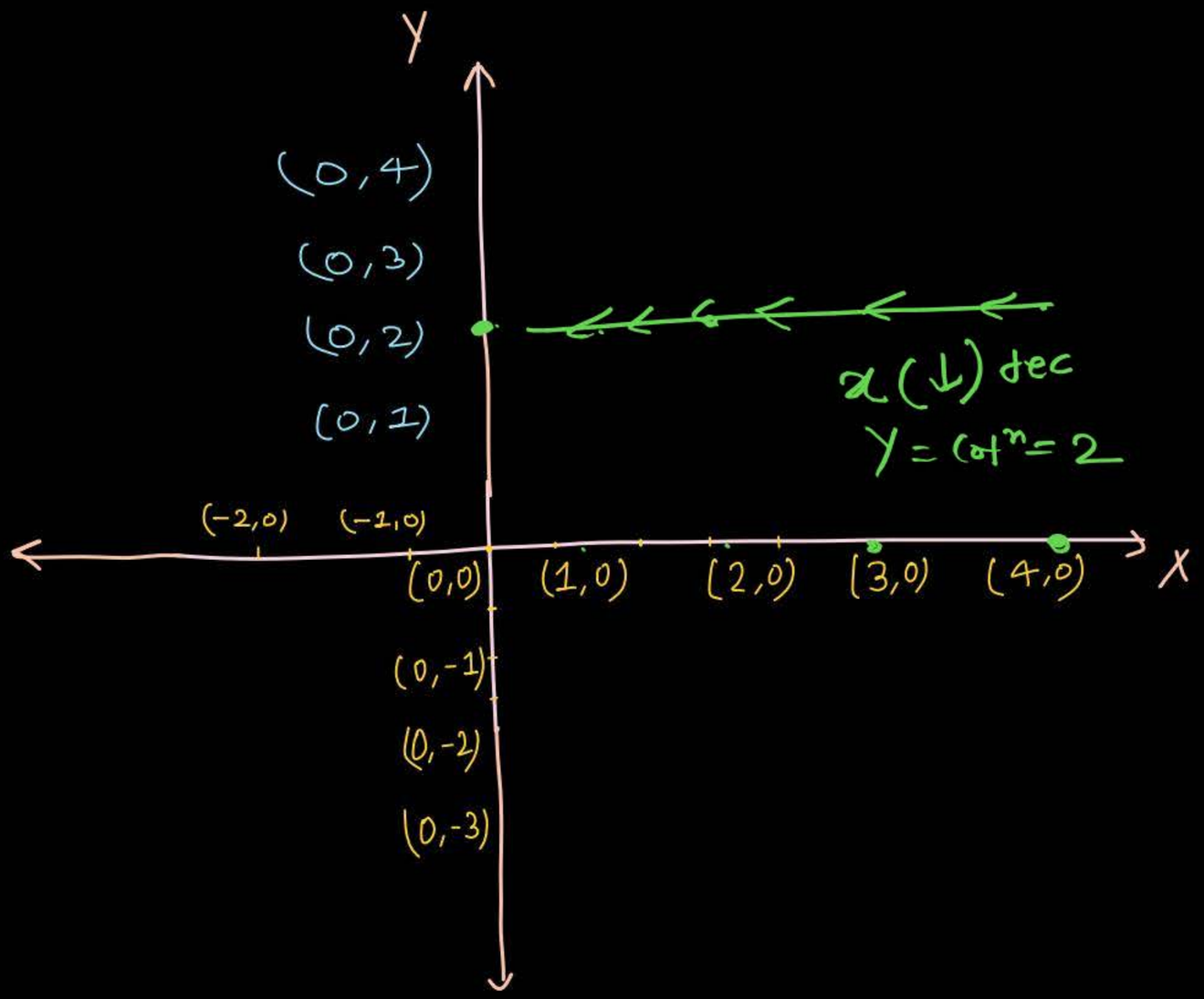
$P = mv$

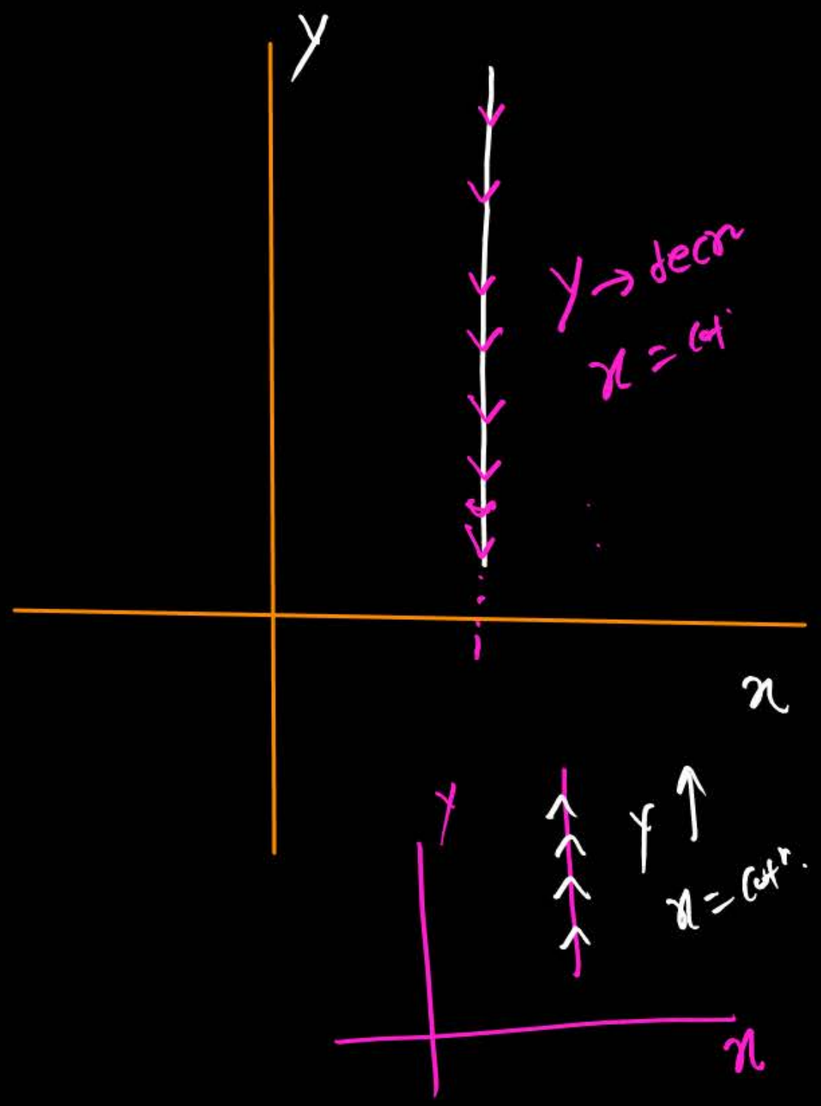
$K.E = \frac{1}{2}mv^2$

MR^ BOY:—
Jis 2-quantity ka graph lena hai usse alawa sare PQ ko constant man lo.

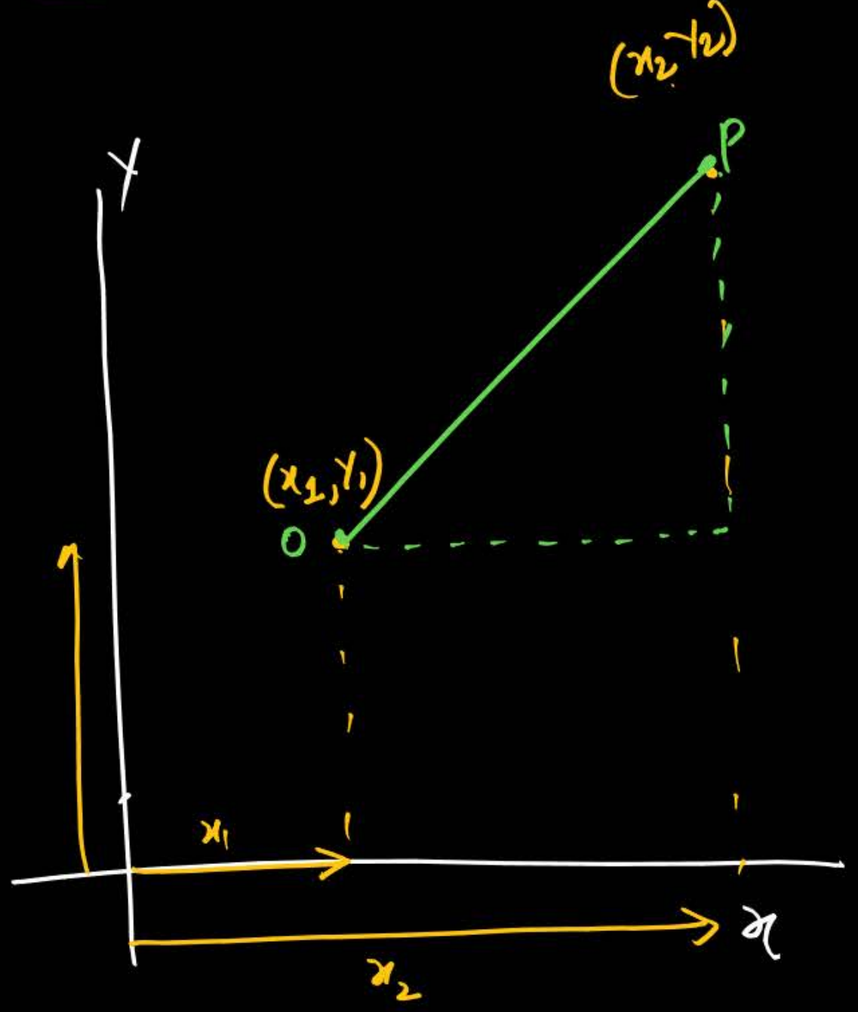


Copy


y
(0,4)
(0,3)
(0,2)
(0,1)
x (↓) dec
y = const = 2
(-2,0)
(-1,0)
(0,0)
(1,0)
(2,0)
(3,0)
(4,0)
x
(0,-1)
(0,-2)
(0,-3)


y
y → decr
x = const
x
y
y ↑
x = const.
x

Copy



OP = ??

HOME Work

↳ 3-4 question in PPT

Sangharsh Assignment -1

& Sangharsh -2 evening me aaega.

Thank

You